

IP-SAKTI Sahayak — Technical Report

Smart India Hackathon, Problem Statement PS-045. Architecture, implementation status, and verification results as of 5 October 2026.

Version	2.0 (replaces v1.1 and v1.2)
Repository	IP_SAKTI_Sahayak (main branch, uncommitted working tree at time of writing)
Status	MVP working on a local machine. Public demo link is a temporary tunnel, not production hosting.
Notice	Smart India Hackathon prototype (PS-045). Not an official Government of India website.

Contents

1. Executive summary
2. Problem and scope
3. System architecture
4. Technology stack
5. Data model
6. Backend API
7. Retrieval, reranking and generation
8. Citation verification and abstention
9. Jurisdiction handling
0. Security and access control
1. Frontend
2. Corpus and data policy
3. Configuration and deployment
4. Testing and verification
5. Known limitations
6. Next steps
7. Appendix: how each number was obtained

1. Executive summary

IP-SAKTI Sahayak is a decision-support system for Ayurveda intellectual property, scientific evidence and regulatory pathways in India, the USA and Australia. It answers only from a curated document corpus, ties each claim to a source passage, checks each claim against that passage, and abstains when evidence is missing or conflicting. It does not determine patentability, approval or efficacy.

As of this report, the following are implemented and running: user and admin areas with separate sign-in, an innovation workflow (profile, classification, evidence, scientific, patents, regulatory, gaps, graph, review), a research assistant with hybrid retrieval and claim verification, an escalation workflow, reports, and an admin console for corpus, monitoring and evaluation.

Verified in this report cycle: 69 of 69 backend tests pass. The frontend type-checks and builds. All 47 page routes load without JavaScript errors for their intended role. Access-control checks return the

expected 401 and 403 responses. Chat, patent search and innovation analysis return real results from the local corpus. The full evaluation suite result is in section 14.6.

2. Problem and scope

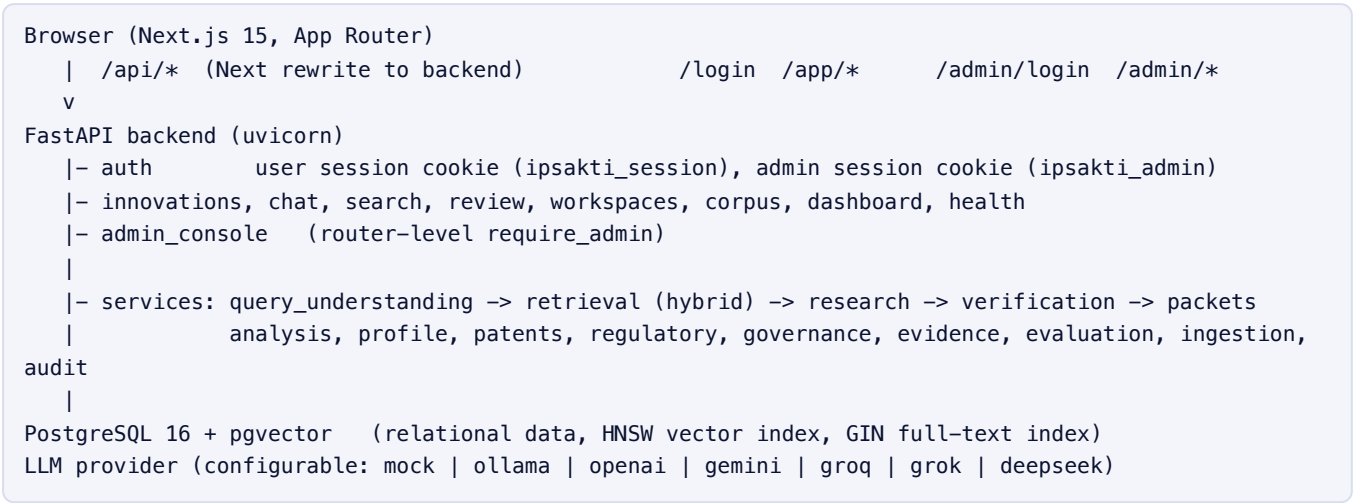
An Ayurveda product needs three answers before it reaches the market: whether it is patentable and what prior art exists, which regulator governs it and under what category, and whether the evidence supports the claims. The sources are statutes, gazette notices, patent databases and journals, often in different languages and naming the same herb differently.

The design goal is traceability and honest abstention. A fluent but unsourced answer is treated as a failure, not as a success.

Out of scope

- Legal advice, patentability opinions, regulatory approval, or efficacy determination.
- Access to TKDL records or other restricted traditional-knowledge databases. The system declares this limit and does not scrape them.
- Live feeds from official sources. The corpus is a curated seed.

3. System architecture



The frontend does not hold any server secret. The browser calls `/api/*`, which Next.js rewrites to the backend. Session tokens are httpOnly cookies that JavaScript cannot read.

4. Technology stack

Layer	Component and version	Notes
Frontend framework	Next.js 15.5.27 (App Router), React 19, TypeScript 5.7	15.1.6 was upgraded because of a known vulnerability that blocked hosting deployment.
Styling and UI	Tailwind CSS 3.4, lucide-react 0.475, framer-motion 13, @xyflow/react 12	Design tokens are defined in <code>tailwind.config.ts</code> . Graph view uses <code>@xyflow/react</code> .

Data fetching	TanStack React Query 5	Admin and user session state are cached and invalidated on sign-in and sign-out.
Fonts	@fontsource: Inter, Noto Sans (with Devanagari), Sora	Self-hosted. The app makes no Google Fonts requests.
Backend framework	FastAPI (≥ 0.110), Pydantic v2, pydantic-settings, uvicorn	Typed request and response models. OpenAPI schema is generated at <code>/docs</code> .
ORM and migrations	SQLAlchemy 2.x, psycopg 3, Alembic	4 migration revisions (section 5.2).
Database	PostgreSQL 16 with the pgvector extension (<code>pgvector/pgvector:pg16</code> in docker-compose)	Vector search (HNSW), full-text search (GIN) and relational data in one database.
Authentication	bcrypt password hashing, PyJWT tokens, httpOnly cookies, CSRF header check	Two token scopes: <code>user</code> and <code>admin</code> .
Testing	pytest 8, pytest-asyncio; Playwright (Chrome) for browser checks	69 backend tests (section 14).
PDF generation	Playwright printing from HTML	Used for this report and the MVP overview.

5. Data model

5.1 Entities (29 ORM classes, 10 enums)

Area	Classes
Identity and tenancy	User, Organization, Workspace, WorkspaceMember
Innovation workflow	Innovation, InnovationProfile, InnovationFeature, Term, PatentFeatureMatch, EvidenceGap, RegulatoryPathway, ScientificStudy
Corpus	Source, Document, DocumentChunk, Patent, IngestionJob
Assistant and evidence	Conversation, Message, RetrievalResult, Evidence, Claim, ClaimEvidence, Feedback
Review and governance	Escalation, Report, AuditLog, EvaluationQuestion, EvaluationRun

Enums: UserRole, WorkspaceRole, Confidentiality, InnovationStatus, FeatureType, RetrievalMethod, SupportStatus, Severity, EscalationStatus, EscalationType.

5.2 Migrations (4 revisions)

Revision file	Purpose
87c58741447b_initial_domain_schema.py	Initial domain schema
06db756477da_innovation_analysis_state.py	Stored analysis state for innovations
4310f8724217_escalation_assignment_and_notes.py	Escalation assignment and notes
8623fabb51aa_feedback_and_source_curation.py	Feedback and source curation

6. Backend API

The live OpenAPI schema lists **107 operations** across **91 paths**. Of the operations, 102 are under `/api`, 4 are health checks, and 1 is the root.

Module	Responsibility
<code>auth.py</code>	User login, register, logout, session and me; separate admin login, logout and session
<code>innovations.py</code>	Innovation CRUD, profile generation and editing, evidence gaps, patents, regulatory, classification, graph
<code>chat.py</code>	Assistant conversations and messages; the full RAG and verification pipeline
<code>search.py</code>	Hybrid search, patent search, scientific search, regulatory search, citation verification, classification
<code>review.py</code>	Escalations, review packets, evidence packets
<code>workspaces.py</code>	Workspace membership and switching
<code>corpus.py</code>	Document upload, coverage matrix, source listing
<code>dashboard.py</code>	User dashboard figures
<code>admin_console.py</code>	Admin-only: users, workspaces, sources, documents and ingestion, RAG and citation monitoring, evaluations, metrics, audit log. Guarded at router level.
<code>health.py</code>	Health and LLM provider status

Rate limits apply to the `chat`, `search` and `ingest` buckets, configured in settings. Errors use a common envelope: `{"success": false, "error": {"code", "message", "details"}}`.

7. Retrieval, reranking and generation

- Query understanding.** Language detection (English and Hindi), terminology normalisation for Ayurveda names, and intent classification.
- Hybrid retrieval.** pgvector HNSW similarity search and PostgreSQL full-text search run together. Results are filtered by jurisdiction, document type and workspace in SQL, before reranking.
- Deterministic reranking.** Each candidate is scored from jurisdiction match (weight 0.12), source authority (weight 0.12) and freshness (weight 0.06), plus relevance terms. There is no learned cross-encoder.
- Authority tiers.** Tier 1 (official law or regulator) = 1.0, tier 2 = 0.85, tier 3 = 0.7, tier 4 = 0.5, tier 5 (user-provided, unverified) = 0.3.
- Freshness labels.** Current = 1.0, Recently checked = 0.85, Potentially stale = 0.5, Unknown = 0.4, Superseded = 0.1.
- Generation.** The selected LLM provider writes an answer from the retrieved passages. In mock mode, the answer quotes retrieved passages directly. Retrieved text is treated as data, not as instructions.

Embeddings: the default provider is `hashing` (1024 dimensions, in-process, no network). The evaluation run in section 14.6 reports the embedding provider as `hashing-ngram`. OpenAI and bge-m3 are available by configuration. Hashing captures shared words, not synonyms, so synonym recall is weaker than with a semantic model.

8. Citation verification and abstention

Each claim in an answer is checked against the cited passage. Verification verdicts are **SUPPORTED**, **PARTIALLY_SUPPORTED**, **UNSUPPORTED** and **CONTRADICTED**. The evidence status shown with an answer can also be **CONFLICTING** or **INSUFFICIENT**.

A confidence score is shown with a signal breakdown: claim verification score, share of fully supported claims, source authority, jurisdiction match and other signals. The score is a heuristic, and the response marks it as such.

The assistant abstains when: the jurisdiction is not supported (rejected at the schema level, see section 9), clarification is needed, the evidence is insufficient, or the query looks like a prompt-injection attempt.

9. Jurisdiction handling

- Supported jurisdictions are **IN**, **US** and **AU**. Request schemas reject any other value with HTTP 422 (verified with `JP`).
- Document jurisdictions also allow **INTERNATIONAL**. Retrieval always includes international documents alongside the selected jurisdiction.
- An innovation's target markets limit its patent matching. An EP-jurisdiction patent is therefore never matched to an innovation.
- Jurisdiction is chosen once on the landing page and kept in the browser. The dashboard, assistant and research pages read the same value.
- Admin analytics group RAG and citation events by jurisdiction. Events without a jurisdiction (for example, scientific queries) are shown under `?`, not hidden.

10. Security and access control

Control	Implementation
Password storage	bcrypt hashing
Session tokens	PyJWT tokens with scope claim. User tokens are in <code>ipsakti_session</code> , admin tokens in <code>ipsakti_admin</code> . Both are httpOnly. The admin cookie uses SameSite=strict.
Scope separation	The user dependency accepts only user-scoped tokens. The admin dependency accepts only admin-scoped tokens and re-checks role=ADMIN on every request.
Cross-scope responses	A user session on an admin endpoint returns 403 (<code>ADMIN_ACCESS_DENIED</code>). An admin session on a user endpoint returns 401.
CSRF	Cookie-authenticated writes must send <code>X-Requested-With: ipsakti</code> . Requests without it return 403 <code>CSRF_CHECK_FAILED</code> .
Frontend route guards	<code>frontend/middleware.ts</code> redirects <code>/app/*</code> without a user cookie to <code>/login</code> , and <code>/admin/*</code> without an admin cookie to <code>/admin/login</code> . Middleware checks cookie presence only. The backend validates the token on every API call.
Admin discoverability	The admin sign-in page is not linked from any public page or from the user app. It carries only the label "Administrator access".
Audit	Key actions are written to the audit log. IP addresses are stored as keyed SHA-256 hashes (keyed with the session secret), not in raw form.

Workspace isolation	Innovations, conversations, documents, reports and escalations are filtered by workspace membership on every query.
---------------------	---

Development defaults. `JWT_SECRET` defaults to a development value. The config raises an error in production if it is not set. Demo accounts use a shared, seeded password for local use only.

11. Frontend

Area	Count and content
Page routes	47 (verified from the build manifest): 3 public (/ , /login , /register), 29 user app routes under /app , 15 admin routes under /admin including /admin/login
Shared components	6 component files: UI primitives and layout, evidence cards, auth form, back navigation
Languages	English and Hindi (lib/i18n.ts)
Size	About 6,300 lines of TypeScript and TSX across app/ , components/ and lib/
Accessibility	Text size control (100%, 112.5%, 125%) saved in the browser. High-contrast mode. Skip-to-content link. Visible focus rings. The hero fits one screen at each text size. A full WCAG 2.0 AA audit has not been done.
Landing page	Sticky accessibility bar, rotating hero (3-second interval, paused on hover, focus and reduced-motion), feature and persona sections, pipeline diagram, sourced industry updates, FAQ and footer. The footer and header carry the prototype notice.

12. Corpus and data policy

The corpus was checked on the running instance (section 17). It contains **40 documents** from **14 sources**.

Dimension	Distribution
Jurisdiction	IN 18, INTERNATIONAL 10, US 7, AU 5
Review status	SEED_SUMMARY 30, DEMO_FICTIONAL 9, PENDING_REVIEW 1
Authority tier	Tier 1: 22, Tier 2: 2, Tier 3: 6, Tier 4: 9, Tier 5: 1
Document type (largest groups)	LEGISLATION 9, PATENT 7, STUDY 6, GUIDANCE 5, RULES 5, REGULATION 4
Update status	Up to date 37, Review due 1, Overdue 1, Superseded 1

- **SEED_SUMMARY**: short excerpts or summaries of real public instruments. These must be checked against the official text before use.
- **DEMO_FICTIONAL**: fictional patents, guidance and the demo innovation. Labelled so nobody mistakes them for real filings.
- **PENDING_REVIEW**: user uploads. Private to the workspace until reviewed.

13. Configuration and deployment

Setting	Default in code and what runs locally
---------	---------------------------------------

LLM_PROVIDER	Code default <code>mock</code> . The local instance reports <code>ollama</code> (local model, no network cost). Providers without an API key fall back to mock with a warning.
EMBEDDING_PROVIDER	<code>hashing</code>
JWT_SECRET	Development default, must be replaced for production
Database	PostgreSQL 16 + pgvector (docker-compose, or a local instance)
Frontend	Production build served on port 3000. <code>/api/*</code> rewrites to the backend on port 8000.

Hosting status

- **Local:** backend (uvicorn, port 8000) and frontend (Next.js production server, port 3000) run on a development machine.
- **Public demo:** a Cloudflare quick tunnel exposes port 3000. The URL changes on every restart, and it stops when the machine sleeps or the tunnel process ends.
- **Planned:** frontend on Vercel, backend and database on a paid platform. Railway was paused because the trial expired. Production hosting has not been set up.

14. Testing and verification

14.1 Backend automated tests

Command: `.venv/bin/python -m pytest -q` from the repository root.

Test file	Tests	What it covers
<code>test_auth_security.py</code>	15	Access-rules matrix: anonymous, user and admin on each route group, cookie scope separation, CSRF, login errors
<code>test_rag.py</code>	16	Assistant pipeline, jurisdiction filtering and mismatch handling, abstention, clarification, refusal
<code>test_retrieval.py</code>	10	Hybrid retrieval, filters, reranking behaviour
<code>test_governance.py</code>	9	Coverage matrix, evidence graph, governance outputs, admin monitoring endpoints
<code>test_jurisdiction_patents.py</code>	9	Patent matching limited to target markets, jurisdiction filter applied in SQL
<code>test_llm_mode.py</code>	7	LLM provider selection, fallback to mock, mock-mode answers
<code>test_integration_flow.py</code>	3	End-to-end flows across user actions
Total	69	

Result: 69 passed in 8.30 seconds before the ingest fix, and re-run after it (section 14.5a).

14.2 Frontend type check and build

Check	Result	Note
<code>npx tsc --noEmit</code>	pass	No type errors
<code>npm run build</code> (Next.js 15.5.27)	pass	Production build with middleware (34 kB). Static and dynamic routes generated.

`npx next lint` (ESLint 8, eslint-config-next 15.5.27, core-web-vitals)

pass

No warnings or errors. Three unescaped-quote errors were fixed during setup.

14.3 Route smoke test (browser)

Each route was opened in a Chrome session for the matching role: anonymous for public pages, the demo user for user pages, and the demo admin for admin pages. Dynamic routes used a real innovation id where one exists, and `x` otherwise. Both status and JavaScript errors were recorded.

Group	Routes	HTTP 200	Page JS errors	Redirects
Public	4	4	0	None (<code>/admin/login</code> is not redirected while signed out)
User app (including 3 public pages in the same run)	32	32	0	None; the user session stays on each route
Admin	14	14	0	None; the admin session stays on each route

Limit of this test: it confirms that each page loads and renders without a JavaScript error for the intended role. It does not check every button or every data state. Dynamic routes were sampled with one id each, and the research route was sampled with one kind.

14.4 Sign-in split and routing (browser)

#	Scenario	Observed result	Status
1	<code>/login</code> has no admin references and no tab	No "admin" text in the DOM, no tab elements	pass
2	User signs in	Lands on <code>/app</code>	pass
3	User opens <code>/admin/dashboard</code>	Redirected to <code>/app</code> . <code>/api/admin/metrics</code> returns 403.	pass
4	Fresh <code>/admin/login</code>	Only admin fields, "Administrator access" label, no demo button, no link to the user page	pass
5	Admin signs in	Lands on <code>/admin/dashboard</code>	pass
6	Admin opens <code>/app</code>	Redirected to <code>/login</code> . Sessions are separate; the admin session does not open the user app.	pass (decision)
7	Admin logs out, then opens <code>/admin/dashboard</code>	Redirected to <code>/admin/login</code>	pass
8	User session opens <code>/admin/login</code>	Redirected to <code>/app</code> (by design). A user must sign out before an admin can sign in from the same browser.	pass (decision)

14.5 API checks (direct requests)

Check	Result	Detail
-------	--------	--------

Anonymous request to a user endpoint	401	Expected
User session on an admin endpoint	403	/api/admin/metrics
Admin session on a user endpoint	401	/api/innovations : admin token is not accepted for user scope
Cookie write without X-Requested-With	403	CSRF_CHECK_FAILED
Workspace members list	pass	Platform admin accounts are not listed in workspace member lists (API filter). Demo list shows the researcher, reviewer and demo user.
Assistant question, jurisdiction IN	200	Answer with 6 evidence items, evidence status SUPPORTED, confidence HIGH (0.97, heuristic), generation mode llm (Ollama)
Assistant question, jurisdiction JP	422	Rejected by schema: input must be IN, US or AU
Patent search, jurisdiction IN	200	Returns DEMO patent records with publication numbers and applicants
Innovation evidence gaps	200	9 gaps for the seeded demo innovation
Innovation patents, regulatory	200	Both return data
Admin RAG monitoring	200	Retrieval statistics
Admin metrics	200	Latency and cost over 3 sampled queries: p50 34.3 s, p95 49.2 s (local Ollama on this machine); estimated cost \$0.00087 per query under the stated assumptions

The latency figure is from three sampled queries on a development machine. It is not a production benchmark.

14.5a Additional functional checks

Browser checks (Chrome) and API checks run after the sign-in split. They do not call the LLM, so they could run while the evaluation suite was in progress.

Check	Result	Detail
"Use demo account" fills user@ipsakti.demo and Demo@12345; Explore Demo link pre-fills the same	pass	Sign-in page button and landing link. The user then signs in manually.
Wrong password shows an error	pass	"Email or password is incorrect." No admin hint
Hindi switch changes the hero and persists after reload	pass	Language stored in the browser
A+ text size sets root font size 125% and persists	pass	Accessibility control
High contrast toggle sets data-contrast=high	pass	Accessibility control
No horizontal overflow at 390 px width on /, /login, /register, /admin/login	pass	Overflow 0 px on each page
Logout ends the user session; /app then redirects to /login	pass	Browser test

Create innovation with a valid wizard	pass	Profile generation and analysis not triggered in this test
Create innovation with empty basic section	pass	422 validation error
Get, patch and classify an innovation	pass	200 responses
Create escalation; user updates status; admin lists escalations	pass	201 on create, 200 on update and list
Create report and export Markdown	pass	200 on both
User cannot read admin RAG or ingest documents	pass	403 on each
Write without X-Requested-With; anonymous list request	pass	403 and 401
Document ingest with jurisdiction JP	pass	422, schema validation
Document ingest with an unknown source id	404 after fix	Found as a 500 by this test. The failed-job record used a source id that does not exist, so the database rejected it. Fix: write the failed-job record only when the source exists. Re-test after backend restart: 404.

Test residue: the checks created test records in the demo database (innovations named "QA Test ...", escalations, and reports). They are labelled and can be removed from the admin console or the API.

14.6 Full evaluation suite

The admin console runs an evaluation of 20 questions across the categories listed below against the live pipeline. Each run writes a record that can be reviewed in the admin console.

Categories (count in brackets): synonym (2), simple factual, multi-source, Hindi, patent, scientific, regulatory, cross-jurisdiction, ambiguous classification, conflicting sources, outdated sources, insufficient evidence, restricted TKDL, prompt injection, medical boundary, legal certainty, misleading citation, Hindi synonym, regulatory Australia.

Run ID 7b1164bc, created 4 October 2026 20:07 UTC (5 October, 01:37 IST). LLM: ollama:qwen2.5:7b (local). Embeddings: hashing-ngram. Top-k = 6.

14.6.1 Retrieval and answer quality (20 questions)

Metric	Value	Meaning
Recall at k=6	0.978	Share of expected sources found in the top 6
Retrieval precision	0.638	Share of retrieved sources that were expected
nDCG	0.877	Ranking quality
Keyword coverage	1.000	Expected terms present in answers
Citation correctness	1.000	Cited sources are the expected ones
Citation entailment	0.794	Share of claims whose cited passage supports them
Unsupported claim rate	0.033	Lower is better
Abstention accuracy	1.000	Correct decisions to answer or abstain

Jurisdiction accuracy	1.000	Answers for the correct jurisdiction
Jurisdiction contamination	0.000	Answers that used out-of-jurisdiction sources
Freshness score	0.990	Harness freshness score for the answers (higher is better)
Harness pass count	20 / 20	Questions the harness marked as passed

Evidence status across the 20 questions: SUPPORTED 11, PARTIALLY_SUPPORTED 5, INSUFFICIENT 3, CONFLICTING 1.

14.6.2 Comparison with baselines (same corpus and questions)

System	Recall at k	Jurisdiction contamination	Abstention accuracy	Citation verification
IP-SAKTI (this system)	0.978	0.000	1.000	Yes, claim-level
Keyword only (BM25)	0.822	0.222	0.850	None
Same retrieval, always answers	n/a	n/a	0.850	Not stated in the run
LLM only, no retrieval	n/a	n/a	0.800	0 of 20 answers had verifiable citations

14.6.3 Latency (this run)

p50 54.3 s, p95 86.2 s, mean 48.2 s per question, on a local machine running a 7B model on CPU. These numbers are much higher than the 34 s p50 in section 14.5, which came from three queries. Production latency depends on hardware and the chosen model.

14.6.4 Innovation classification (8 cases, 13 labels across 3 jurisdictions)

Agreement with the expected label: 13 of 13 (100%).

Read this with care. The expected labels were assigned by the project team against the seeded pathway definitions. The classifier uses the same definitions. So 100% agreement shows that the logic is consistent with those definitions, not that it matches expert regulatory judgement. The labels need review by a qualified regulatory expert before this number is used as evidence of accuracy.

14.6.5 What these results do and do not show

- They show that, on this seed corpus and these 20 questions, retrieval finds the expected sources, abstention and jurisdiction rules behave as designed, and citation checks catch unsupported claims at a low rate.
- They do not show accuracy on real documents. The corpus is a seed of 40 documents, and 9 of them are fictional (section 12).
- The question set is small (20) and written by the project team, so it can favour the system's own design.
- The harness counts 20 of 20 questions as passed under its own rules. This is not an independent expert review.

15. Known limitations

- The corpus is a seed of 40 documents. Statute summaries need checking against official texts. Patents are fictional.
- Hashing embeddings do not match synonyms well. A semantic embedding model is not connected in the current setup. Synonym questions passed, but two of them were only partially supported.
- Classification labels were written by the project team and need expert review (section 14.6.4).
- Latency depends on the local model. The evaluation run shows p50 of 54 s; an earlier sample of three queries showed 34 s.
- Rate limiting runs in-process. Background jobs run inside the web worker. A larger deployment needs Redis and a separate worker.
- The public demo link is temporary and changes on restart. Production hosting is not set up.
- No formal WCAG 2.0 AA audit has been done.
- Admin access is held by the project lead for now. Further admin accounts are not yet needed.

16. Next steps

1. Add further administrator accounts only when the team needs them, using the create-admin command.
2. Decide production hosting and a real contact address.
3. Connect a semantic embedding model and an LLM key, then re-run the evaluation suite.
4. Verify the corpus against official texts and move reviewed documents out of SEED_SUMMARY.
5. Run a formal accessibility audit.

17. Appendix: how each number was obtained

Number	Source
107 API operations, 91 paths	Live <code>/openapi.json</code> from the running backend
29 ORM classes, 10 enums	Class definitions in <code>backend/app/models/orm.py</code>
4 migrations	<code>backend/alembic/versions/</code>
10 endpoint modules, 17 service modules	<code>backend/app/api/endpoints/</code> and <code>backend/app/services/</code> (excluding <code>__init__.py</code>)
69 tests	<code>pytest --co</code> and the pass run in section 14.1
47 page routes	<code>frontend/.next/app-path-routes-manifest.json</code> after the production build, minus <code>/_not-found</code> and <code>/icon.svg</code>
About 8,400 backend and 6,300 frontend lines	<code>wc -l</code> over <code>backend/app</code> and <code>frontend/app</code> , <code>components</code> , <code>lib</code>
40 documents, 14 sources, review and tier counts	Admin documents and sources endpoints on the running instance
Retrieval weights, tier and freshness values	<code>backend/app/services/retrieval.py</code>